

PROJECT REPORT GUIDELINES-2022

PAPER SIZE, BINDING, MARGINS, FONTS AND LINE SPACINGS

Instructions:

Font: Times New Roman

Font Size:

Chapter Title: 16 Bold (In Capital letters, Centered)

Side Headings: 12 Bold (In Capital letters)

All other text 12 (Normal text)

Line Spacing: 1.5

Paper Size: A4 [One side printing]

Spiral binding: Color: Light Blue

Margins:

Left margin: 1.5 inch

Right margin: 1 inch

Top: 1 inch

Bottom: 1 inch

Number of Pages:

Minimum – 60; Maximum – 100

Header: Project Name (Left) and Page number (Right)

Footer: Department of Computer Science, CHRIST (Deemed to be University)
[with font size 10 for both header and footer; Include border line for header and footer]

PAGE NUMBERING

1. Lower-case Roman numerals (e.g., iii, iv, v) for the front matter/preliminary material from acknowledgements. No page numbering for Table of Contents (TOC).
2. Arabic numbers (e.g., 1, 2, 3, etc.) starts from chapter 1. (This can be achieved by giving a section break)
3. No Header and footer for TOC, Certificate and Title Page.
4. Chapters, Sections, Subsections are numbered as per the following interpretation: example
1.3.2 stands for Chapter 1, section 3, Subsection 2.

IMPORTANT INSTRUCTIONS

1. Do not end a page with a title. Shift the title to the next page.
2. Do not end a page with a single line of a new paragraph, Start the new paragraph in the next page.
3. Do not include Thumb index, transparency sheets etc., within the report.
4. Use a drawing canvas to start drawing the image.
5. Strictly follow the given format for bibliography.
6. Follow the guidelines for Screen shots/Diagrams/Tables.

7. Use automatic table of content option available in documentation tools (MS-Word, Open Office, etc.)

FIGURES AND TABLES

1. Position Figures and Tables as close as possible to the text to which they are referred.
2. Figures and Tables should be numbered in ascending order for each chapter. Example: the second image in chapter 3 will be Fig 3.2.
3. The number of the figure and its caption should be typed immediately below it in Times New Roman Size 10.
4. The number of the table and its caption should be typed immediately above it in Times New Roman Size 10.

ACKNOWLEDGMENTS

1. Guide
2. All those who helped to complete the project

ABSTRACT

The abstract should condense the report to 200 words or less in single-space format. Do not include any sub-title or bullets in abstract. The abstract should contain:

- a. The most essential background information
- b. The purpose and scope of the project
- c. Major outcomes and recommendations

CONCLUSION

Rewrite the aim and objective of the project with a brief note on the major modules. Summarize the available features, results, and limitations. Conclude with your future enhancements.

REFERENCES PAGE INSTRUCTIONS

1. In the body of your text, refer to the source of your information by inserting consecutive numbers in brackets at the end of each segment of cited information—like this [1]. These reference numbers can also be inserted within a sentence [2], without changing punctuation. You may also cite your reference in your text as follows: “As Smithsky [3] points out,..

2. Unless you are referring to a complete book or article, you will also need to identify the page number(s) of your source of information. Indicate exact page numbers of a source within your brackets after a comma [4, pp. 3-6], or by a simple rhetorical device in your text such as "However, on page 79 of [5] the author seems to contradict himself when he states. . .". If you must refer to more than one separate page or source in the same reference, use semicolons for separation: [6, p. 46; 7, pp. 29-31; 9, pp. 8-12].
3. Once you have numbered a source, use the same number for all subsequent references to that source. References at the end of quotation marks "are punctuated with the period after the reference" [8, p. 23].
4. Always use square brackets around reference numbers [10, pp. 78-85] to distinguish from equation numbers, which are given in parentheses (6).
5. Single-space individual references, with no indentation for a second or third line. Double space between separate references.
6. Use a common abbreviation for a Journal title if there is one, e.g., IEEE Electron Device Lett. Otherwise give the full name of the journal.
7. End each entry with a period.
8. List each source only once on your reference page.

Books Format:

Author's last name, first name. *Book title*. Edition and other additional information. City of publication: Publisher, year.

- [1] Allen, Thomas B. *Vanishing Wildlife of North America*. Washington, D.C.: National Geographic Society, 1974.
- [2] Hyde, Margaret, and Elizabeth Forsythe. *Suicide: The Hidden Epidemic*. New York: Watts, 1986.
- [3] Baron, Alvin. *Bud's Easy Research Paper Computer Manual For IBM PC's*. 3rd ed. New York: Lawrence House Publishers, 2000.

Magazine & Newspaper Articles Format:

Author's last name, first name. "Article title." *Periodical title* Volume # Date: inclusive pages.

- [1] Kanfer, Stefan. "Heard Any Good Books Lately?" *Time* 113 21 July 1986: 71-72.

Website or Webpage Format:

Author's last name, first name (if available). "Title of work within a project or database." *Title of site, project, or database*. Editor (if available). Electronic publication information (Date of publication or of the latest update, and name of any sponsoring institution or organization). Date of access and <full URL>.

Note: If you cannot find some of this information, cite the available information.

- [1] Devitt, Terry. "Lightning injures four at music festival." *The Why? Files*. 2 Aug. 2001. 23 Jan. 2002 <<http://whyfiles.org/137lightning/index.html>>.
- [2] Dove, Rita. "Lady Freedom among Us." *The Electronic Text Center*. Ed. David Seaman. 1998. Alderman Lib., U of Virginia. 19 June 1998 <<http://etext.lib.virginia.edu/subjects/afam.html>>.

Sample reference page:

- [1] Allen, Thomas B. *Vanishing Wildlife of North America*. Washington, D.C.: National Geographic Society, 1974.
- [2] Boorstin, Daniel J. *The Creators: A History of the Heroes of the Imagination*. New York: Random, 1992.
- [3] Hall, Donald, ed. *The Oxford Book of American Literacy Anecdotes*. New York: Oxford UP, 1981.
- [4] Kanfer, Stefan. "Heard Any Good Books Lately?" *Time* 113 21 July 1986: 71-72.
- [5] Devitt, Terry. "Lightning injures four at music festival." *The Why? Files*. 2 Aug. 2001. 23 Jan. 2002 <<http://whyfiles.org/137lightning/index.html>>.

Appendix

Use the appendices to avoid interrupting the body of the report with excessive detail, thereby improving readability. Number the appendices with a structure like: A, A.1, B, etc. The appendices might include, but are not limited to:

- a. A "User's Manual" or instructions for operating and maintaining the system you've designed.
- b. Additional charts, graphs and tables.
- c. Mathematical derivations, proofs, etc.
- d. Detailed calculations (theoretical developments, error and sensitivity analyses, etc.)
- e. Budget status.
- f. Gantt chart and/or other planning information.
- g. An analysis of the "design concepts" topics (e.g. health and safety) given separately. Indicate how your design is affected by each topic.

List of Figures

Fig. No.	Figure Name	Page No.
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List of Tables

Table. No.	Title	Page No.
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List of Abbreviations

Example:

UNICEF United Nations Children's Fund

Strictly follow the Order of Pages given below

Title Page (Refer the attached template)

Certificate page (Refer the attached template)

Acknowledgments (Thank all those people whose efforts have contributed towards the successful completion of your project)

Abstract

Table of Contents

List of Tables

List of Figures

Abbreviations (optional)

All Chapters (Refer the sample TOC given below)

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References

Explanation of Contents of Various Chapters**Chapter 1: Introduction**

Begin this chapter with a summary of the contents of this chapter (Maximum two lines).

The Introduction comprises of several sections as described below:

- 1.1 **Project Description:** A Good summary of the background and context of the project. Explain the issues/limitations identified in the existing system/product and the need to resolve the issues.
- 1.2 **Existing System:** Summarize existing work in the area related to your project.
- 1.3 **Objectives:** Aims and objectives of your project to be written in a very concise manner. Clearly define what you are going to achieve in your project (in not more than 40 words).
- 1.4 **Purpose, Scope and Applicability:**
 - 1.4.1 **Purpose:** Explain (i) why you are doing this project? (ii) what is the significance of this project work (iii) How and in what all ways will this project improve the current system?
 - 1.4.2 **Scope:** (i) Give a brief overview of the assumptions made, limitations of the project and the methodology adopted (ii) Discuss the major issues addressed by your project (iii) Explain the main functions of your project.
 - 1.4.3 **Applicability:** Discuss the applications (both direct and indirect) of your project and its contributions to society.
- 1.5 **Overview of the report:** What will be the contents covered in the report and how is it arranged into various chapters?

Chapter 2: System Analysis and Requirements

Begin this chapter with a summary of the contents of this chapter (Maximum two lines).

2.1 Problem Definition: Define and explain the problem on which you are working in the project. Provide the details of the complete problem and then divide the problem into sub problems.

2.2 Requirements Specification: Define the requirements of the system without focusing on how they will be achieved. Requirements can be identified from the limitations and problems of the existing system

2.3 Block Diagram: Draw a diagram of the system under study, illustrating the major components and functions.

2.4 System Requirements

2.4.1 User Characteristics: Different types of users who will be using and dealing with your system.

2.4.2 Software and Hardware Requirements:

(i) Software Requirements — List (a) the Operating System (b) The Compiler (c) Testing Tools (d) Linker and (e) Libraries (f) Packages (g) Virtualization Software etc. required to compile, link and install the software

(ii) Hardware Requirements--- List (a) Processor (b) Graphics Card (c) Disk capacity (d) RAM capacity etc. required to run the software.

2.4.3 Constraints: Explain Hardware Limitations, Language Requirements, Interfaces to other Applications, Reliability Requirements, Regulatory Policies, Platform Dependencies or any other constraints related to your system.

2.5 Conceptual Models: Now that you have understood the problem domain and the requirements, produce a model of the system, reflecting the operations/functions that can be performed on/by the system and the allowable sequence of these operations. Conceptual Models can include Context Diagram, Data Flow Diagrams (At least 0 and 1 Level), Entity Relationship (ER) Diagrams, Class Diagrams, System Flow Charts etc.

Chapter 3 - System Design

Begin this chapter with a summary of the contents of this chapter (Maximum two lines).

This chapter discusses the required features, functionalities and operations in detail. The description should include screen layouts (not screen shots), business rules, process diagrams, pseudocode and other relevant documentation.

3.1 System Architecture

In this section, you can describe the architecture of your system.

E.g., The system under consideration will be based on 2-tier (client-server) architecture, 3-tier architecture (presentation tier, application logic or business logic tier, data tier) etc. Describe the architecture corresponding to your system with a detailed explanation of the components/functionalities. Explanations can be supplemented with appropriate Architecture Diagrams.

3.2 Module Design

While designing the project, follow a divide a conquer approach wherein the overall problem (project) is divided into more structured and manageable parts. Each one of the individual parts or modules are developed separately. Once all the modules are completed, they should be integrated into one complete system.

In this section, you should explain all the modules of your project along with their functionalities.

3.3 Database Design

In this section, you should explain how you have organized, arranged, managed and manipulated the data pertaining to your project. This section should explain the mapping of entities & relationships to tables.

3.3.1 Tables and Relationships: Define the structure of each table and its relationship with other tables in the database.

3.3.2 Data Integrity and Constraints: Define and explain all the validity checks (validations) and constraints you are providing to maintain integrity of data.

3.4 System Configuration (optional)

3.5 Interface and Procedural Design

3.5.1 User Interface Design: Define user, the task being performed, the environment in which task is performed and how you intend to map those requirements in order to develop a “User Interface”. Describe the external and internal components and the architecture of your user interface. Illustrate the interface with the help of some rough pictorial views of the User Interface and its components.

3.5.2 Application Flow/Class Diagram: Using Process flow diagrams, Class diagrams etc. represent the systematical flow of procedure that aids in improving the comprehension of the process and helps the programmer during implementation.

3.6 Reports Design: Define (i) The structure of the reports that will be generated by the system (ii) Inputs given to generate the report (eg. Time period, Name of product etc.) (iii) Output fields in the generated report etc.

NOTE: If your criteria do not match with the above-mentioned designs, then refer various CASE Tools for your application and find a design suitable for your work. Use a design approach appropriate for your project and don't blindly select the given designs just because they are mentioned in the guidelines.

4. Implementation

4.1 Implementation Approaches: Define the plan of implementation and procedure adopted for converting the objectives into an actual working model.

Reference: 1. Roger S. Pressman, Software Engineering, A practitioner's Approach, 7th edition, McGraw-Hill Education, 2010

2. "<http://www.nutcache.com/blog/8-top-project-management-approaches-methods-techniques/>

4.2 Coding Standard: Define the coding standards used for writing the code of your project.

4.3 Coding Details: Include only the important portions of code (applets code, forms code etc.) The lines of code should be commented appropriately for explaining the purpose of a piece of code. Screen shot of the output of the program code can be included to demonstrate the functionality of the code.

4.3 Screen Shots: Input Screen shots & Output Screen shots.

5. Testing

5.1 Test Cases: Test cases are all about "How you are going to test a requirement?" Define the test cases for various test scenarios in your project.

Eg: Test Scenario: "Validate the login functionality of Admin"

This scenario requires three test cases (conditions) as follows

1. Login -successful.
2. Login - unsuccessful when the incorrect password is entered.
3. Login - unsuccessful when the incorrect username is entered.

Explain the purpose of each test and expected output/result.

5.2 Testing Approaches: Explain the approach of testing.

- ii. Unit Testing
- iii. Integration Testing

5.3 Test Reports: Explain the test results and reports based on your test cases, which would prove that your project will produce the expected results under different conditions. Produce the test reports for different test cases by giving different sample inputs and showing the corresponding outputs.

6. Conclusion

6.4. Design and Implementation Issues

Poor design and/or implementation can cause failure or rejection of a software system.

Design Issues: Explain the challenges such as requirements volatility, design process, quality issues (e.g., scalability, performance, security, usability etc.), distributed software development, fast-changing technology etc. faced during the design of your project.

Implementation Issues: Explain the challenges faced by your system during implementation such as operating environment, installation of the system, code conversion, change over, training etc.

References:

<http://apprize.info/usability/engineering/18.html>

6.5. Advantages and Limitations

Advantages: Explain how a problem identified with the existing system has been resolved (ii) Contribution of your work towards improving the performance of the existing system.

Limitations: (i) Explain the functionalities/requirements you were not able implement partially or completely (ii) limitations identified by you during the testing of your project that you were not able to modify (iii) Suggestions for improvement received during demonstration of your project in your company.

6.6. Future Scope of the Project: (i) Discuss the new related areas of investigation identified during the development of the project. (ii) Parts of the current work that could not be completed due to time constraints and/or problems encountered.